

NAME: \_\_\_\_\_ PERIOD: \_\_\_\_\_ DATE: \_\_\_\_\_

## Who Chose Whom?

AP Statistics · Unit 1 · Topic 1.10 · B: 2026-09-25 · E: 2026-10-02

### [STAR] TODAY'S PROBLEM

Suppose a school website posts the question, “Should the cafeteria go cashless?” Any visitor may answer. Of 412 responses, 71% say Yes. That same week, a teacher randomly selects three homerooms and asks every student in those rooms. Of 68 responses, 44% say Yes.

The website headline is “Students back a cashless cafeteria.” The teacher’s headline is “Students are split on going cashless.”

#### Two polls of student opinion

| Poll      | $n$ | Yes | Selection       |
|-----------|-----|-----|-----------------|
| Website   | 412 | 71% | Visitors opt in |
| Homerooms | 68  | 44% | 3 rooms random  |

#### FIRST TAKE — before the video (5 min)

Which headline would you trust more for what students at this school think? Name one detail behind your choice.

#### [BOOK] TRUST STARTS WITH SELECTION (keep on the page)

First name the **statistical question** and the **population** it asks about. A method that uses chance to choose from that population can produce a representative sample. In a **voluntary-response sample**, people choose themselves; people with strong opinions may be overrepresented. A large number of responses reduces random sampling variability, but it does not erase bias from how the sample was selected.

**Finish after the video:**

**DOK 1** (a) State the question of interest and the population the two headlines are trying to describe.

**DOK 2** (b) For each poll, describe how people entered the sample. Name the sampling problem, if any.

**DOK 3** (c) Which percentage should a reader trust more for what students at this school think? Cite the selection feature that settles your choice. Explain why 412 responses do not repair the other poll, and propose one change that would make the website poll more trustworthy.

**Frame:** I would trust \_\_\_\_\_% more because the sample was formed by \_\_\_\_\_.

**Frame:** The 412 responses do not fix \_\_\_\_\_ because \_\_\_\_\_.

**EXIT — turn this in**

One thing the video changed about my first take: \_\_\_\_\_